

SUPPORTING INFORMATION

Multi-Omics Analysis Identifies Paxillin as a Biomarker of Doxorubicin

Resistance via Cytoskeletal Remodeling and Immune Exhaustion

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Supplementary Figures

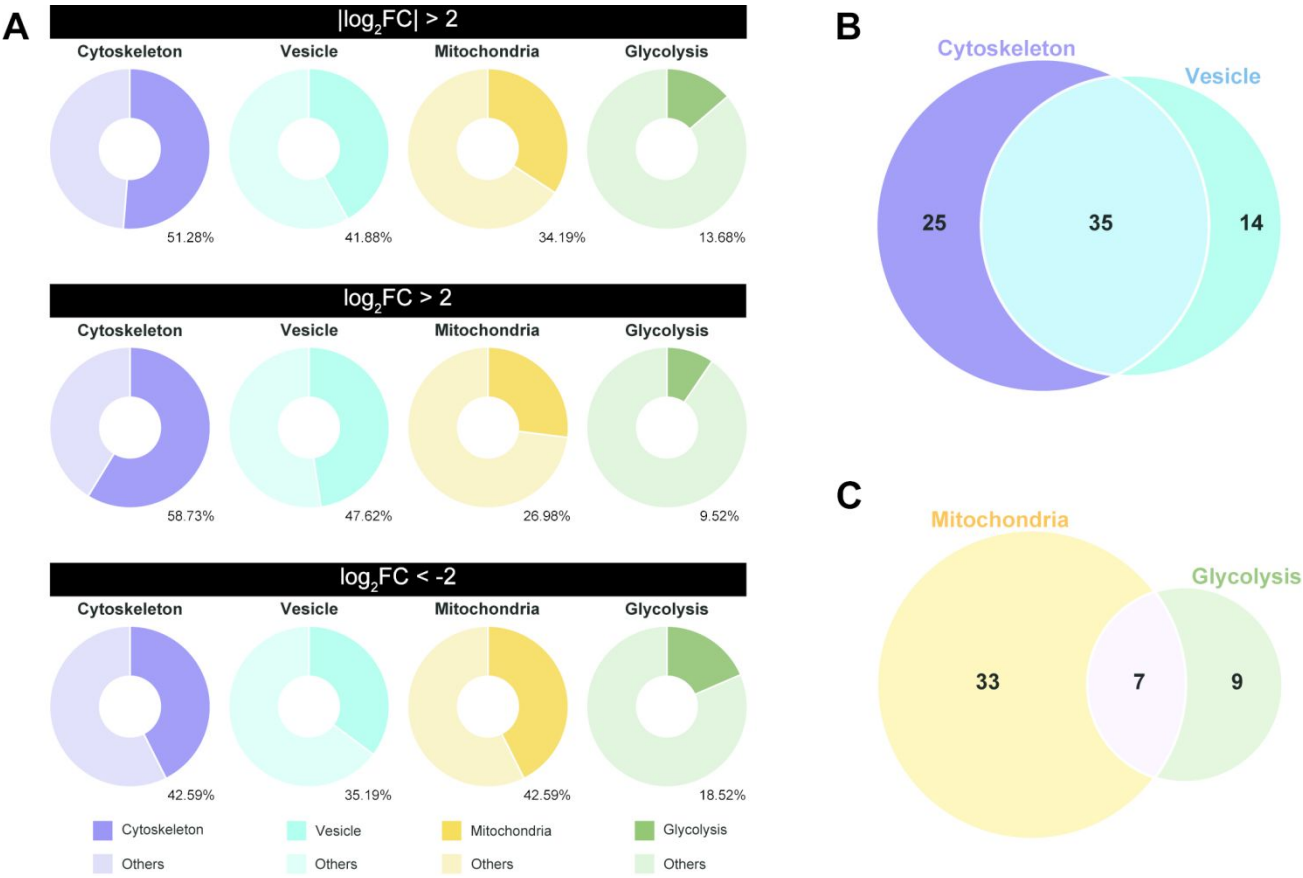


Figure S1. Functional annotation of differentially expressed proteins. (A) Proportion of differentially expressed proteins annotated to Cytoskeleton, Vesicle, Mitochondria, and Glycolysis categories among the total differentially expressed proteins. (B-C) Overlap of annotated protein among differentially expressed proteins ($|\log_2FC| \geq 2$): overlap of Cytoskeleton and Vesicle proteins (B); overlap of Mitochondria and Glycolysis proteins (C).

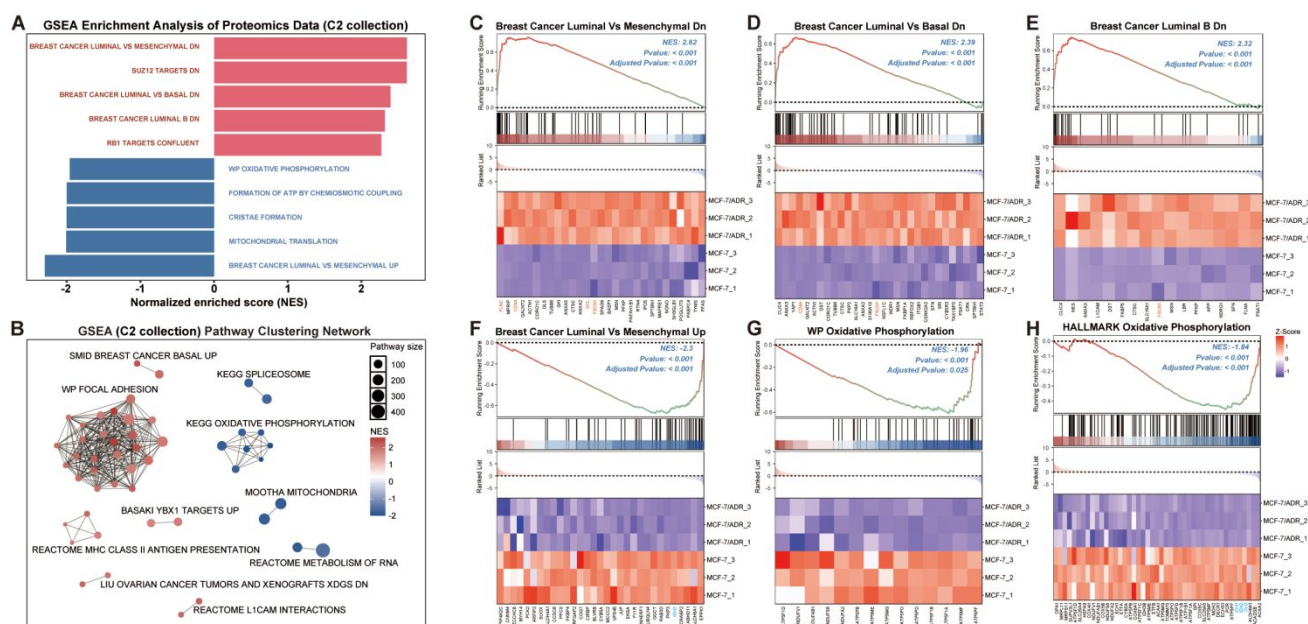


Figure S2. GSEA enrichment using the Molecular Signatures Database (MSigDB). (A) Bar plot of the top 5 pathways from GSEA analysis using curated gene sets (C2). (B) Cluster plot of significantly altered GSEA pathways from MSigDB C2 collection (p adjust < 0.05). (C-F) EMT-related GSEA enrichment curves and heatmaps of core enrichment proteins. (G-H) OXPHOS-related GSEA enrichment curves and heatmaps of core enrichment proteins.

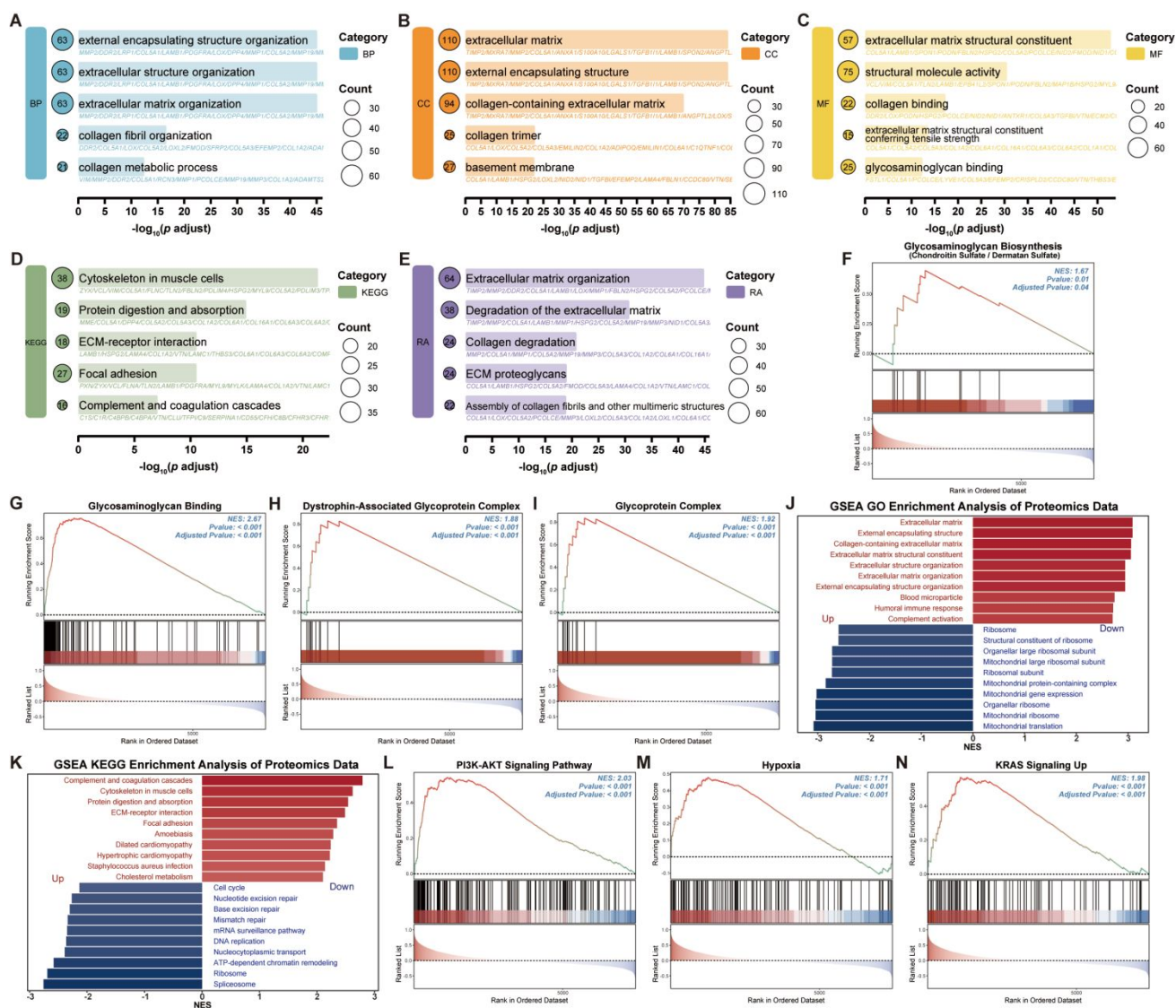


Figure S3. Functional analysis of PXN in the clinical proteome. (A-E) Bubble plots from ORA analysis of differentially expressed proteins ($|\log_2FC| \geq 0.5$, $q < 0.05$) in the FUSCC HER2-low breast cancer cohort. Samples were grouped based on the median PXN protein expression level, and differential protein expression was analyzed using the limma package. (F-I) GSEA enrichment curves of GAG-related pathways in the clinical proteome. (J-K) GSEA bar plots of pathway enrichment. GO gene sets GSEA analysis (J) and KEGG gene sets GSEA analysis (K). (L-N) EMT-related GSEA enrichment curves.

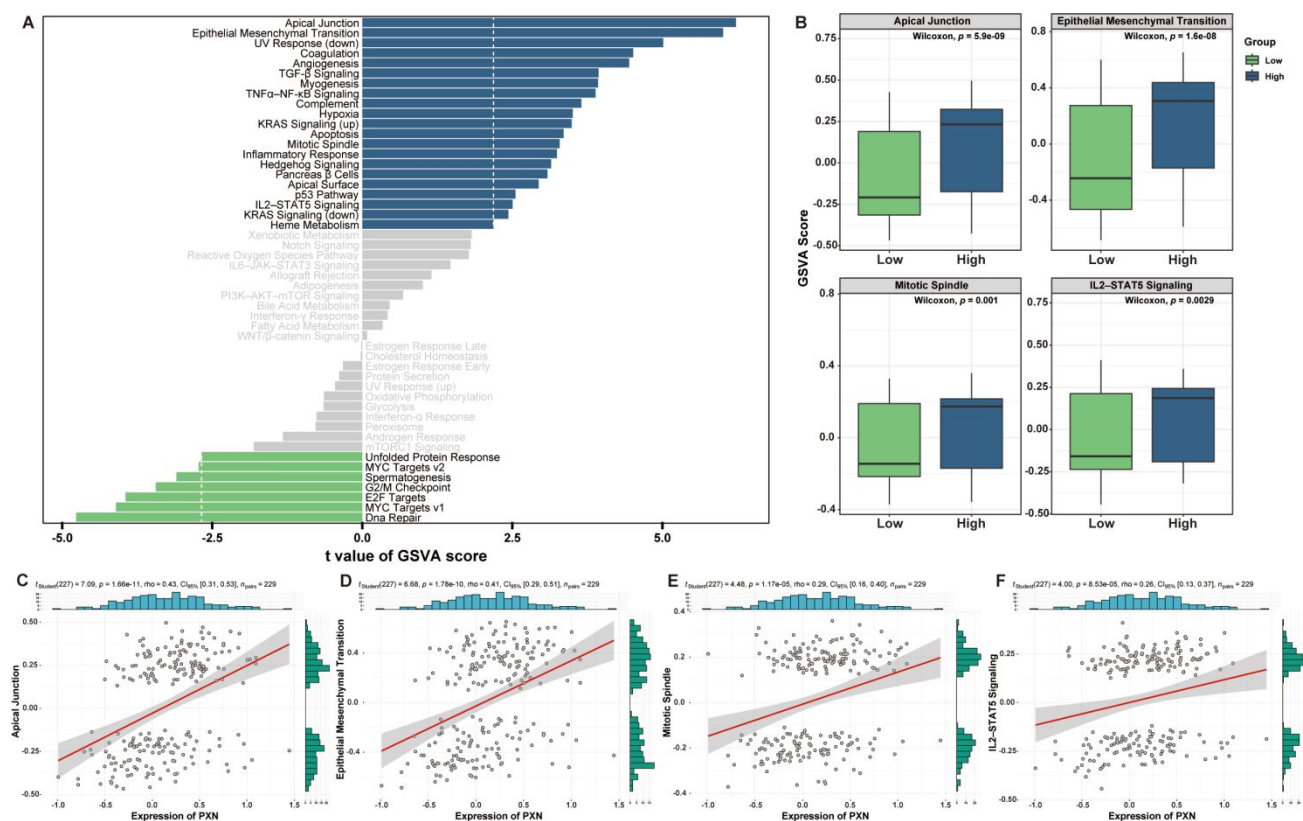


Figure S4. Single-sample GSEA analysis of the clinical proteome. (A) GSEA analysis of the clinical proteome using Hallmark gene sets. Blue and green indicate upregulated and downregulated pathways, respectively. The dashed line represents the threshold for significance. (B) GSEA analysis grouped boxplot, showing inter-group differences across different pathways. Wilcoxon test was used for inter-group comparison. (C-F) PXN pathway correlation analysis, showing the correlation between PXN expression levels and pathway activity.

We clustered enriched pathways and visualized their relationships using aPEAR (R package). Enrichment analysis was filtered at p adjust < 0.05 (Benjamini-Hochberg). Pairwise term similarity was computed with the Jaccard index based on gene-set overlap, and a term-term network was constructed with terms as nodes and Jaccard similarity as edge weights. To reduce visual clutter, weak edges were pruned using outer cutoffs (outerCutoff = 0.4), and only significant terms were kept with inner cutoff (innerCutoff = 0.05). Modules were identified by Markov clustering (MCL). For the main network, node color encoded statistical significance, and node size reflected the number of genes in each term.

Protein-protein interaction (PPI) network construction

PPI networks were constructed using the Search Tool for the Retrieval of Interacting Genes/Proteins (STRING, <https://string-db.org/>) database. The PPI networks were imported into Cytoscape (v3.10.3) for visualization and further analysis. Hub genes were found with the CytoHubba tool using the MCC (Maximal Clique Centrality) method, and the top 10 nodes were reported.

Machine learning for feature proteins selection

Multiple machine learning methods were applied to evaluate associations between selected resistance-associated proteins and patient prognosis. For these analyses, RNA-seq gene expression (TPM) and overall survival (OS) data were retrieved from TCGA-BRCA. To reduce perioperative bias, cases with OS less than 30 days were excluded.

For feature selection and survival modeling, random survival forest (RSF) was trained in randomForestSRC with Surv (OS days, OS event) as the response. The model used 1,000 trees and log-rank splitting. Variable importance (VIMP) scores were computed and used to rank features. Support vector machine recursive feature elimination (SVM-RFE) was implemented with the caret package using the linear SVM (svmLinear), 5-fold cross-validation, and subset sizes (5, 10, 15, 20). The risk label was constructed from survival data. Features were ranked by iteratively removing each variable and quantifying the resulting decrease in cross-validated accuracy, and the final selected subset together with per-feature elimination scores was recorded. Gradient-boosted Cox modeling was trained in xgboost using the survival:cox objective (XGBoost-Cox). Hyperparameters (learning rate, maximum tree depth, minimum child weight, subsample proportion, and column subsampling per tree) were optimized with the mlr3 AutoTuner using 5-fold cross-validation and Harrell's C-index over 30 evaluations. The optimal configuration was retrained for approximately 300 boosting rounds. The model's linear predictor served as the risk score, and feature-importance values were exported. Boruta was applied using the same risk label as in the SVM-RFE analysis. It was run for a maximum of 300 iterations to identify relevant proteins. Tentative features were resolved with the TentativeRoughFix procedure, and confirmed features were retained and ranked by their mean importance.

Survival analysis

We evaluated the association between target protein expression and prognosis in breast cancer by analyzing patients with at least 30 days of follow-up for overall survival (OS) or recurrence-free survival (RFS). Survival analysis was conducted using the Kaplan-Meier method, and the survival curves were visualized using survival and survminer (R packages). Patients were divided into PXN-High and PXN-Low groups based on PXN expression levels, using the best cutoff method with the surv_cutpoint function. All data were sourced from the TCGA dataset and the FUSCC HER2-low breast cancer cohort.

Immune infiltration estimation

The transcriptome sequencing results from the FUSCC HER2-low breast cancer cohort were used to analyze immune infiltration in breast cancer. In this study, patients were divided into two groups based on the median expression of PXN. Subsequently, we applied MCPcounter to estimate the relative abundance of 10 immune cell types in the tumor tissues. CIBERSORT was used to determine the relative proportions of 22 immune cell types using the gene signature matrix (LM22). Gene expression data were input into the CIBERSORT algorithm with 1000 permutations to calculate *p* values for each immune cell type. The xCell method was then employed for a more detailed deconvolution of the breast cancer expression data. The xCellAnalysis function was used to compute immune cell infiltration scores for each sample. Finally, immune score, stromal score, and ESTIMATE score were calculated using the ESTIMATE algorithm.